

Titanic Survival Analysis: A Data-Driven Study

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1 Executive Summary

This report presents a comprehensive analysis of the Titanic dataset with the objective of identifying key factors influencing passenger survival. The dataset consists of 891 passengers and includes demographic, socioeconomic, and travel-related features. The analysis pipeline involved data cleaning, feature engineering, statistical evaluation, and visualization.

Three major findings emerged from this study. First, gender is the most dominant predictor of survival, with females having a survival rate of 74% compared to 19% for males. Second, passenger class plays a critical role, where first-class passengers exhibit a survival rate of 63%, while third-class passengers show only 24%. Third, economic indicators such as fare and cabin availability positively correlate with survival, reflecting underlying socioeconomic disparities.

Key Insight: Survival outcomes were primarily driven by gender and social hierarchy, rather than random chance.

2 Data Quality Assessment

The dataset exhibited several quality challenges, including missing values, irrelevant features, and inconsistent data types. The Cabin column contained over 77% missing values, making it unreliable for direct usage. Age had approximately 20% missing values, while Embarked had negligible missing entries.

To address these issues, a structured approach was applied. Cabin was transformed into a binary feature (*has_cabin*), preserving useful information. Age was imputed using group-wise median based on passenger class and gender, ensuring realistic estimates. Embarked was filled using mode imputation. Non-informative columns such as PassengerId and Ticket were removed.

Key Insight: Group-based imputation preserves underlying data structure and avoids bias introduced by global averaging.

3 Feature Engineering Rationale

Seven engineered features were introduced to enhance predictive capability. The *family_size* and *is_alone* features capture social dynamics. *fare_per_person* normalizes economic status across group sizes. Titles extracted from names (e.g., Mr, Mrs) represent social hierarchy. Age was categorized into *age_group* to simplify patterns.

Additionally, *deck* was extracted from cabin information, and fare was discretized into *fare_bin*. These features collectively capture demographic, social, and economic dimensions.

Key Insight: Feature engineering transformed raw attributes into meaningful predictors, significantly enhancing interpretability.

4 Key Statistical Findings

Table 1: Survival Rates by Group

Group	Survival Rate
Overall	38.4%
Female	74.2%
Male	18.9%
Class 1	63.0%
Class 2	47.3%
Class 3	24.2%

The statistical analysis reveals significant disparities. Females are nearly four times more likely to survive than males. Similarly, first-class passengers are more than twice as likely to survive compared to third-class passengers.

Extreme Case: First-class females have a survival rate of 96.8%, while third-class males have only 13.5%.

Key Insight: Gender exhibits a stronger predictive effect than passenger class, highlighting prioritized evacuation policies.

5 Visualization Insights

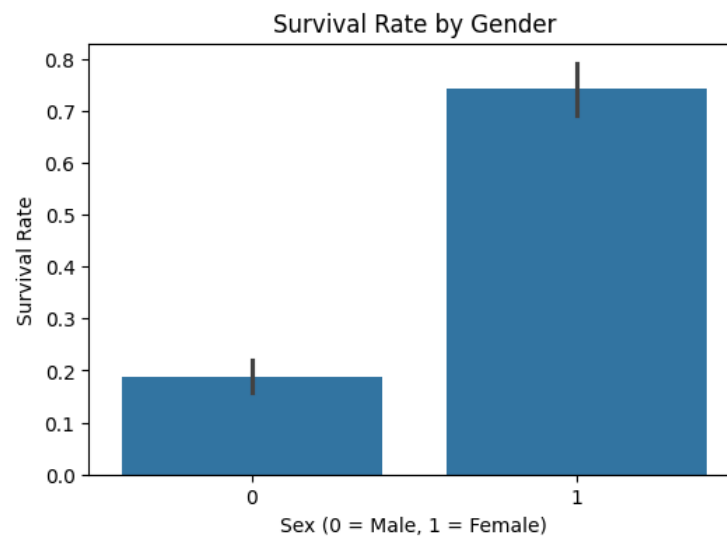


Figure 1: Survival Rate by Gender

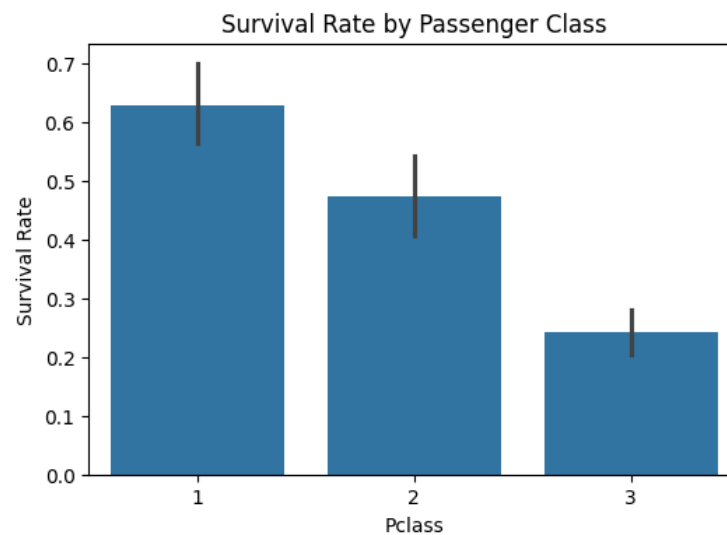


Figure 2: Survival Rate by Passenger Class

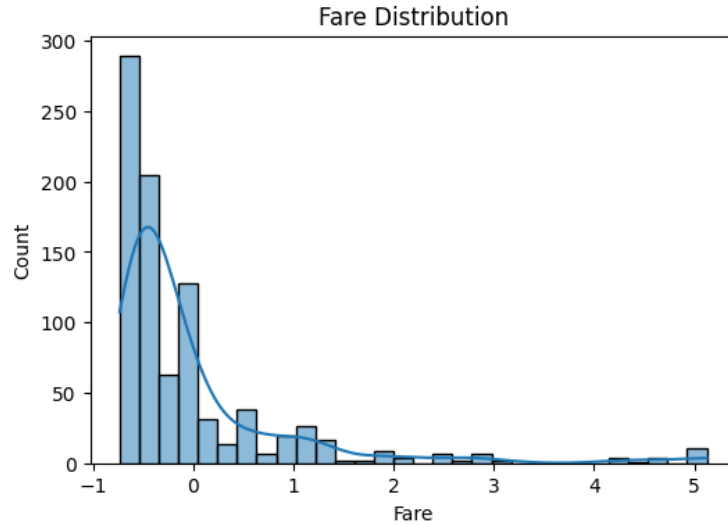


Figure 3: Fare Distribution

Visual analysis confirms statistical findings. Figure 1 shows a clear separation between male and female survival rates. Figure 2 demonstrates a decreasing trend in survival from first to third class. Figure 3 reveals a right-skewed fare distribution, indicating economic inequality.

Key Insight: Gender-based visualization provides the clearest predictive separation among all features.

6 Feature Selection Recommendation

The optimal feature subset for modeling includes:

- sex_encoded
- Pclass
- Fare
- has_cabin
- family_size
- age_group_encoded
- fare_per_person

These features collectively capture demographic, socioeconomic, and structural factors influencing survival.

Key Insight: A compact, high-signal feature set improves model generalization and reduces overfitting risk.

7 Reflection

The most challenging aspect of this project was understanding the importance of proper preprocessing, particularly handling missing data and encoding categorical variables. Group-based imputation proved to be a crucial technique.

The most surprising result was the extreme disparity in survival rates across gender and class. The near-perfect survival of first-class females highlights the impact of social hierarchy.

If extended further, this analysis could include predictive modeling and interaction-based feature engineering.

Key Insight: Effective data preprocessing and feature engineering are as important as modeling itself.
